

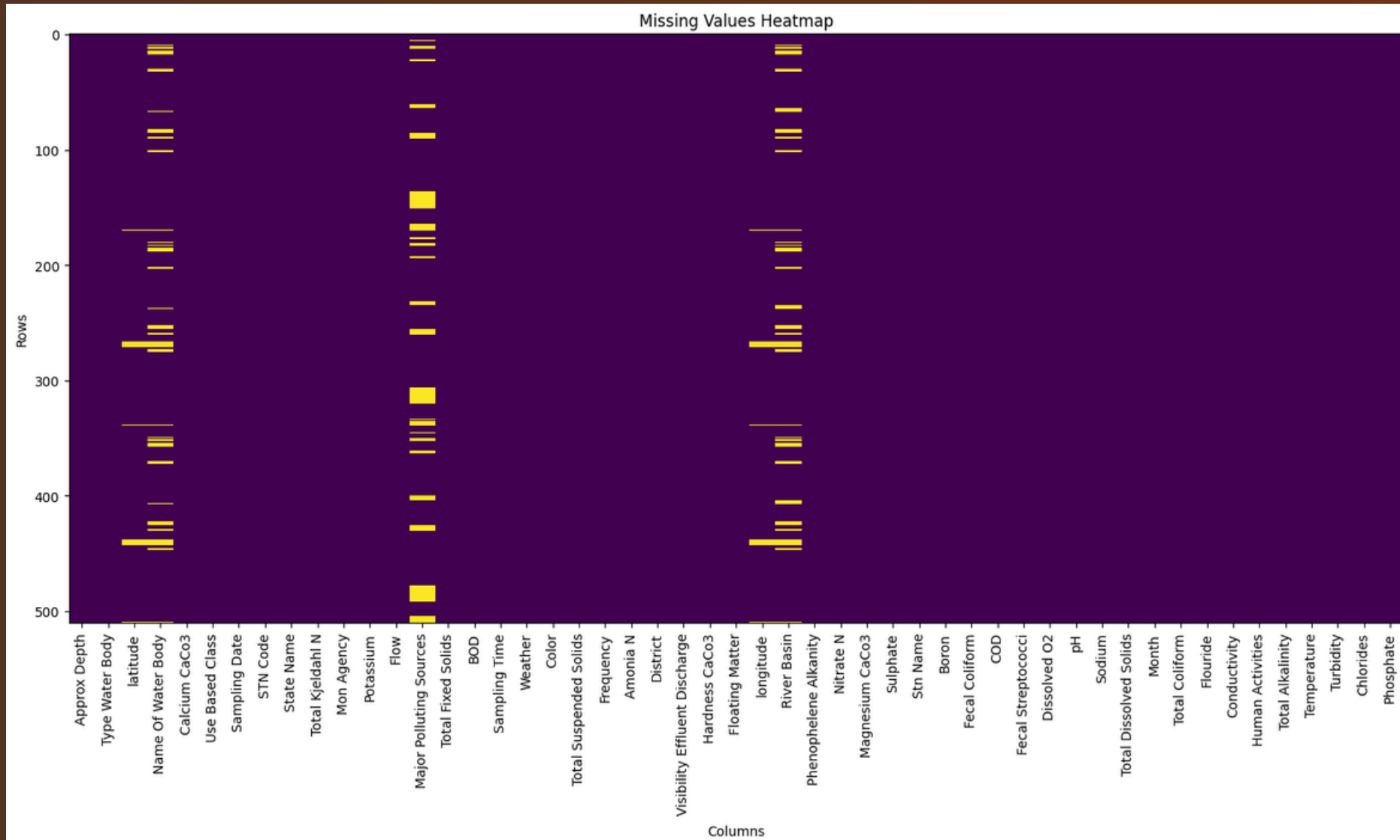
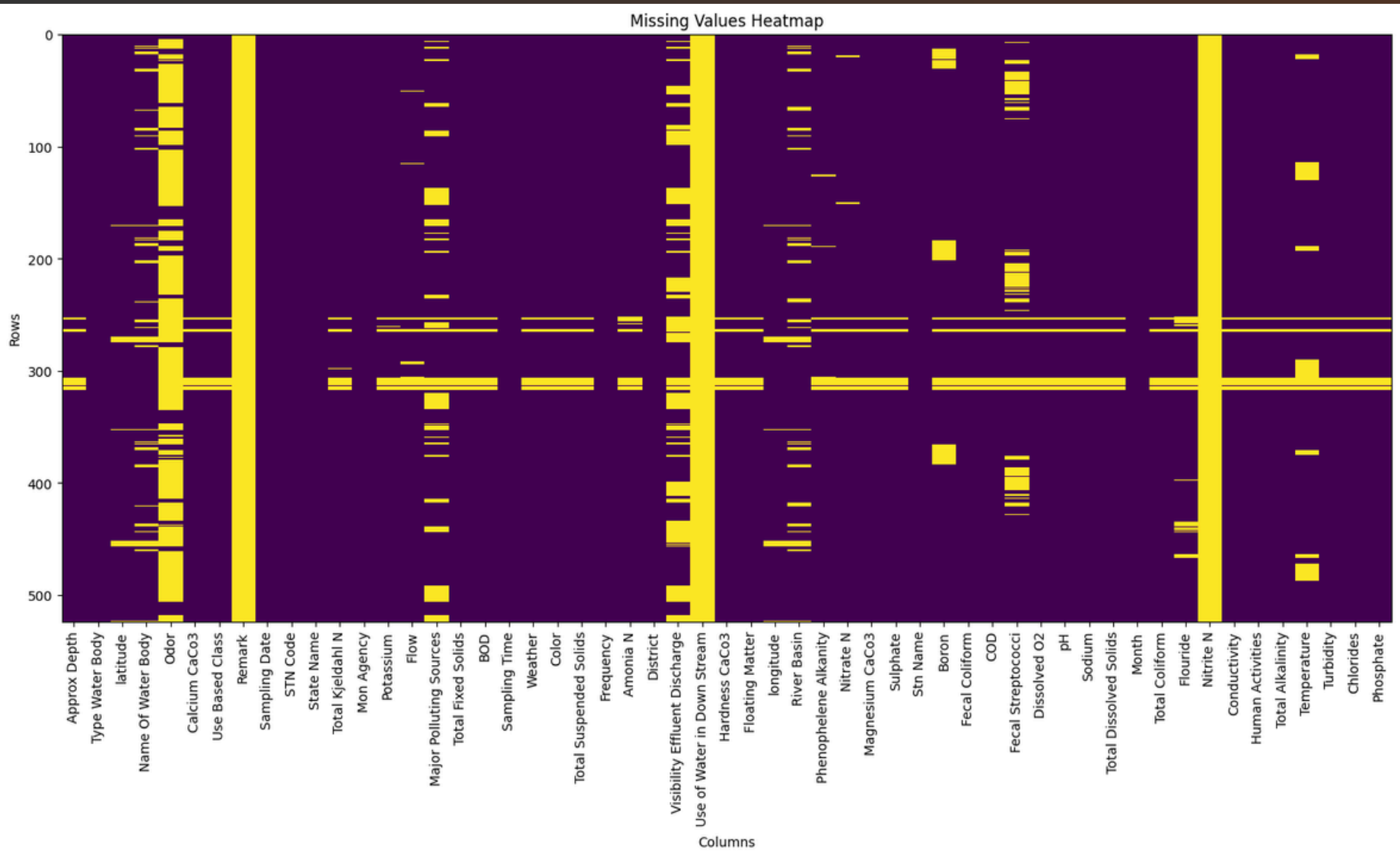
Multi-classification - Water Quality Classification

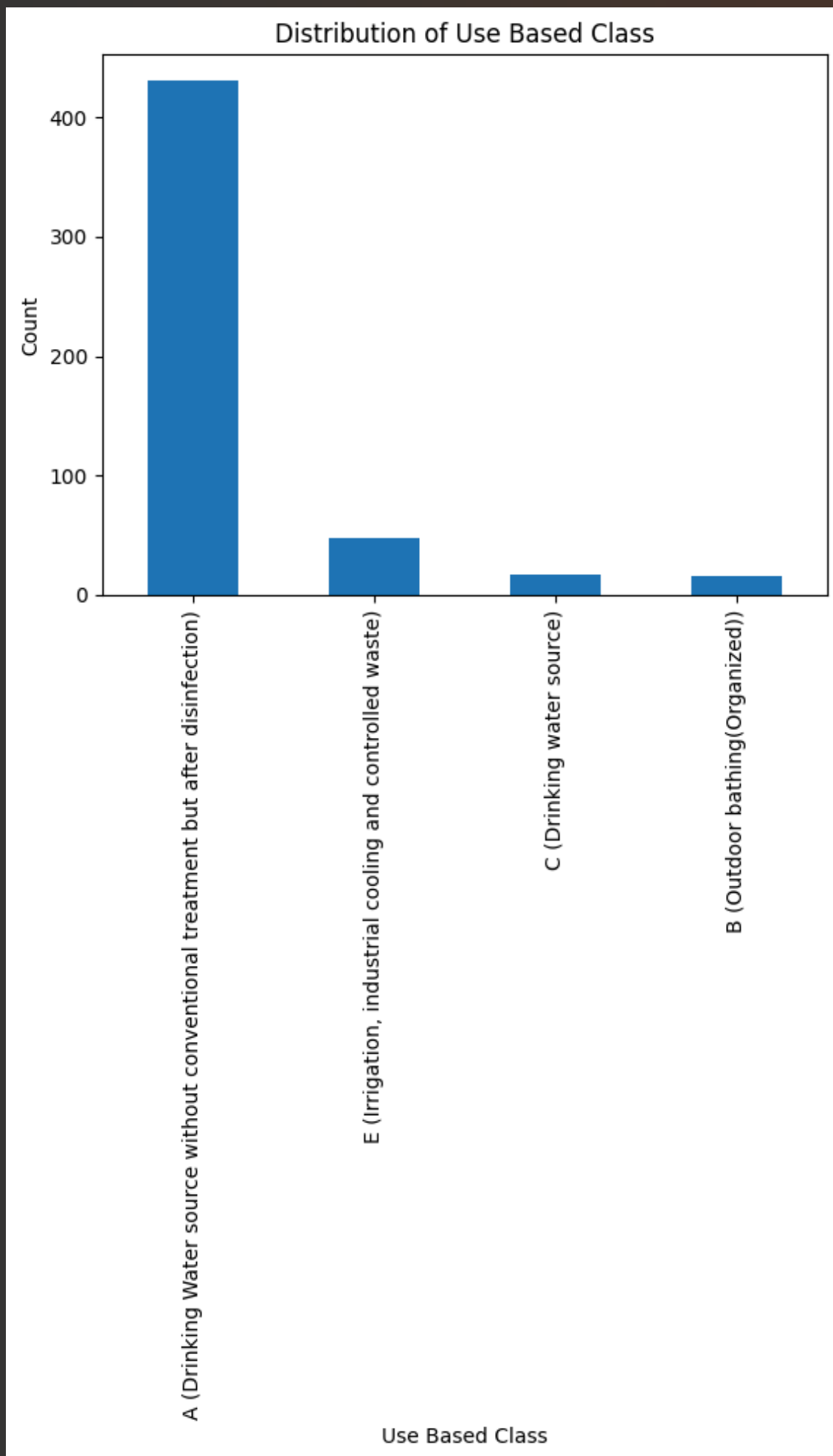
Project Type - Multi-classification

Contribution - Individual

Name - Manoj B

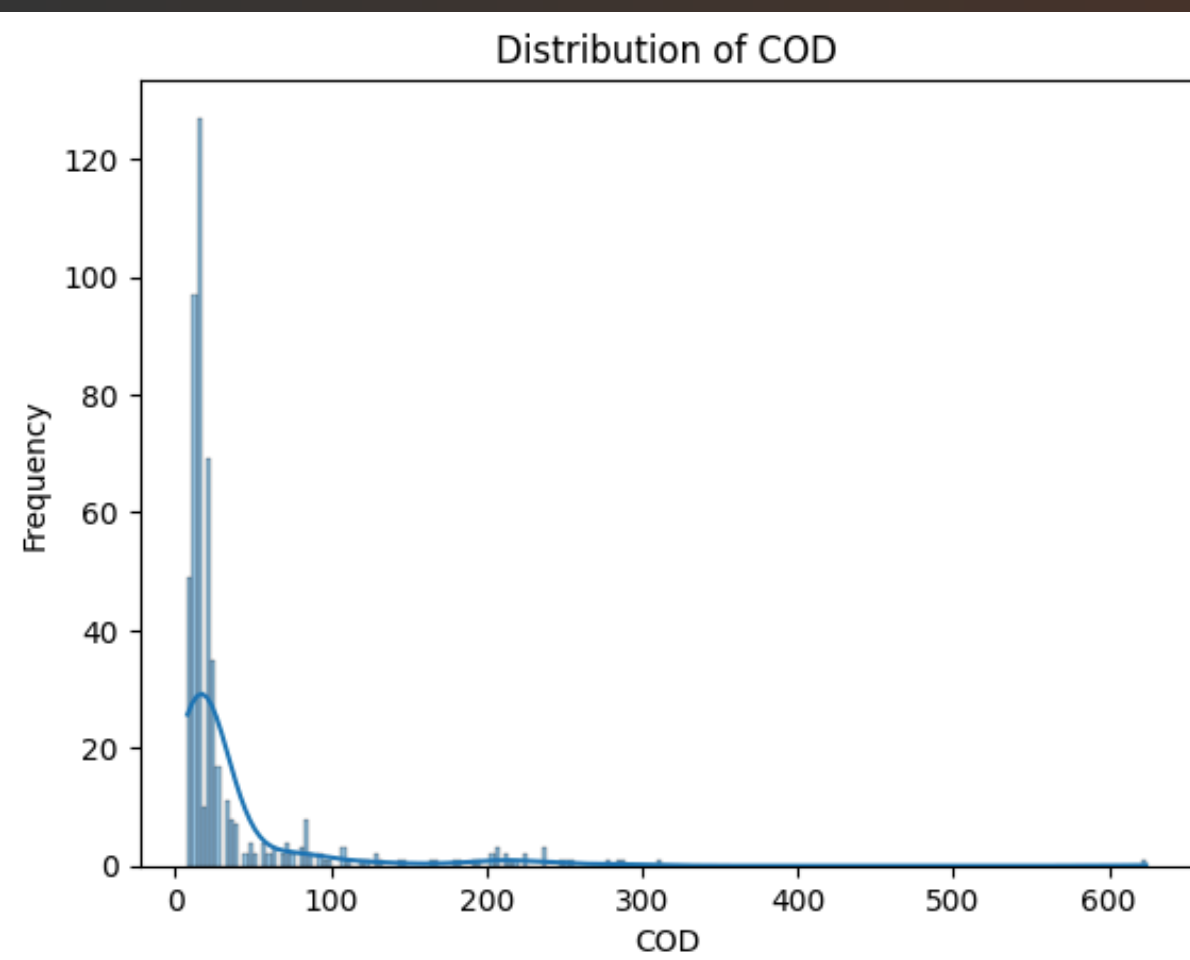
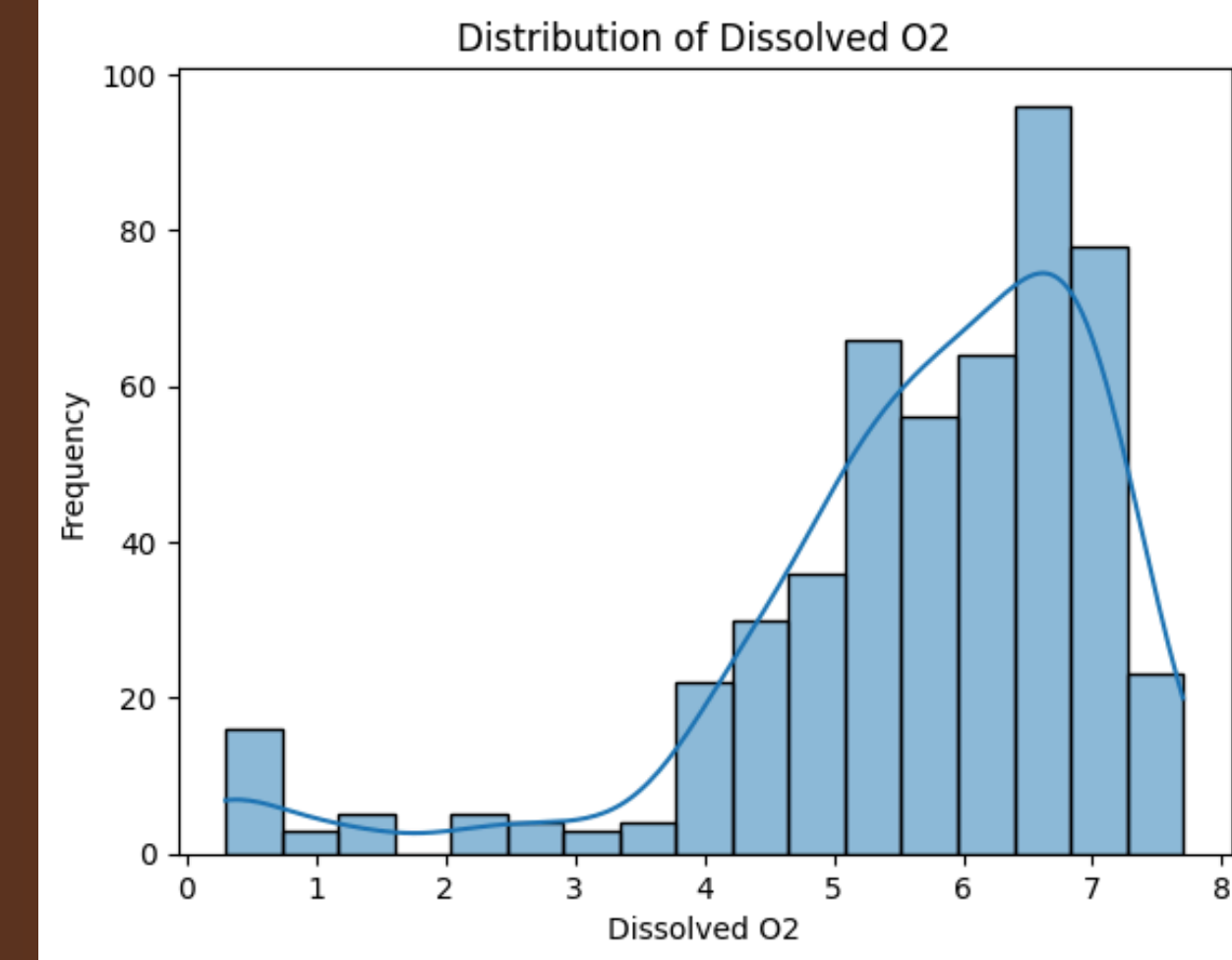
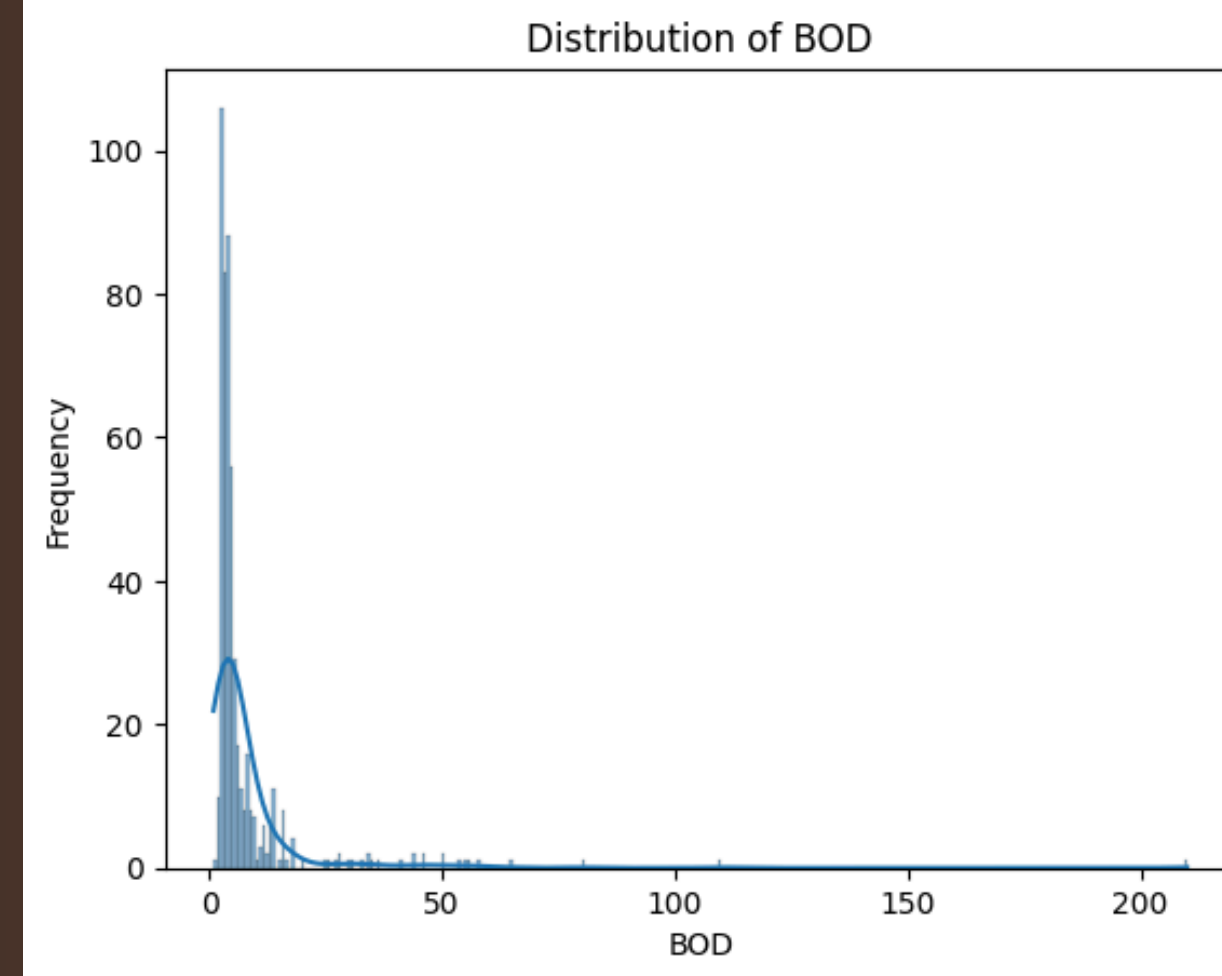
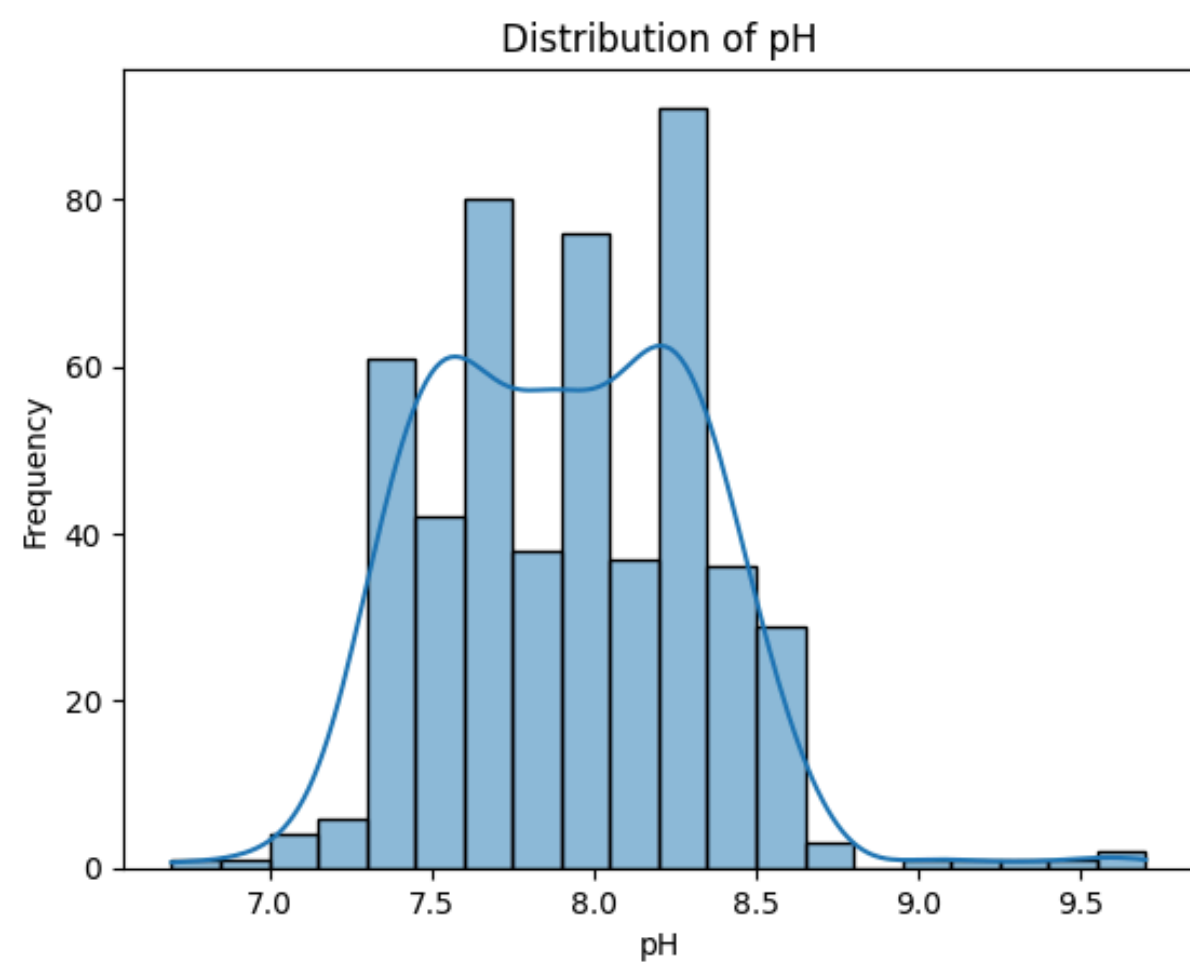
Data Wrangling



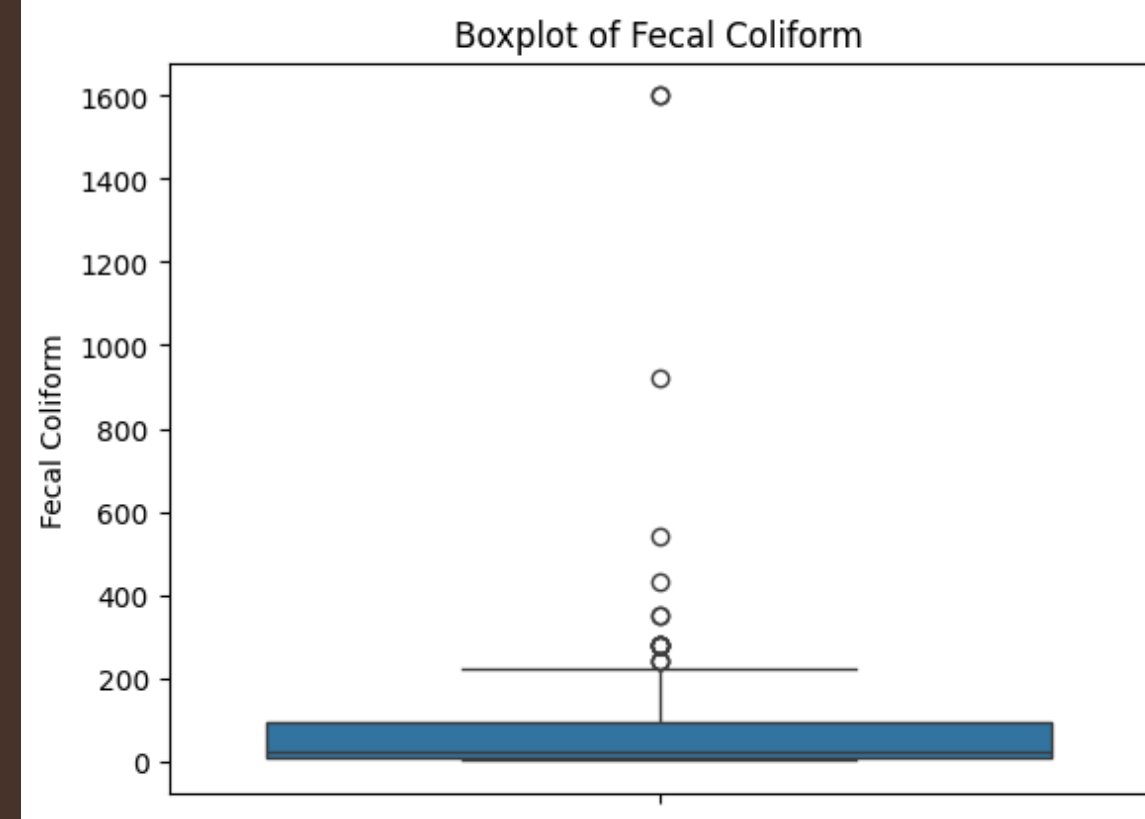
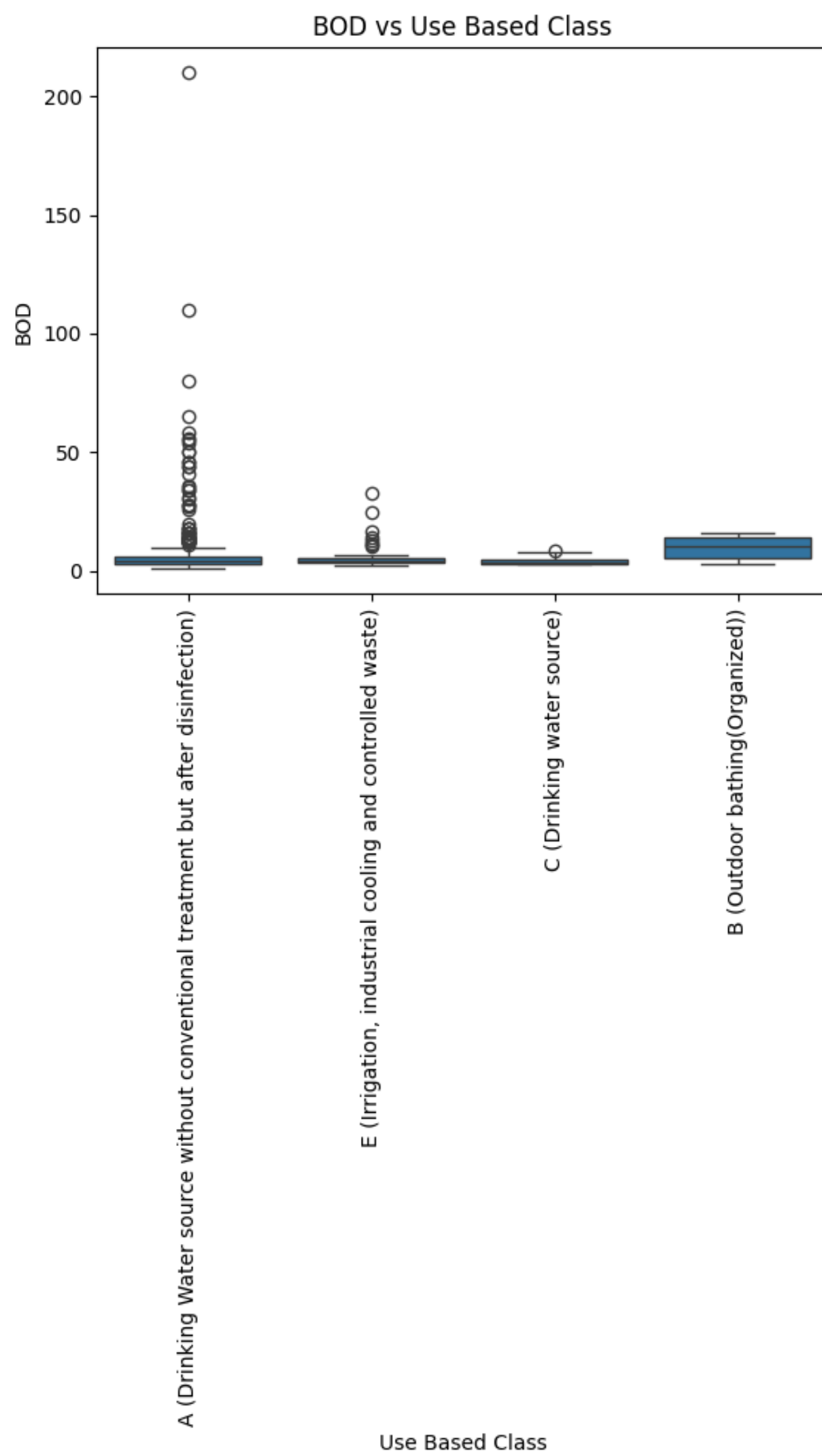


Why to choose this plot: To understand the distribution of target classes and check whether the dataset is imbalanced before training a machine learning model.

What info I got: Most samples belong to Class A (Drinking water source without conventional treatment but after disinfection), while Classes B, C, and E have much fewer records, indicating strong class imbalance.

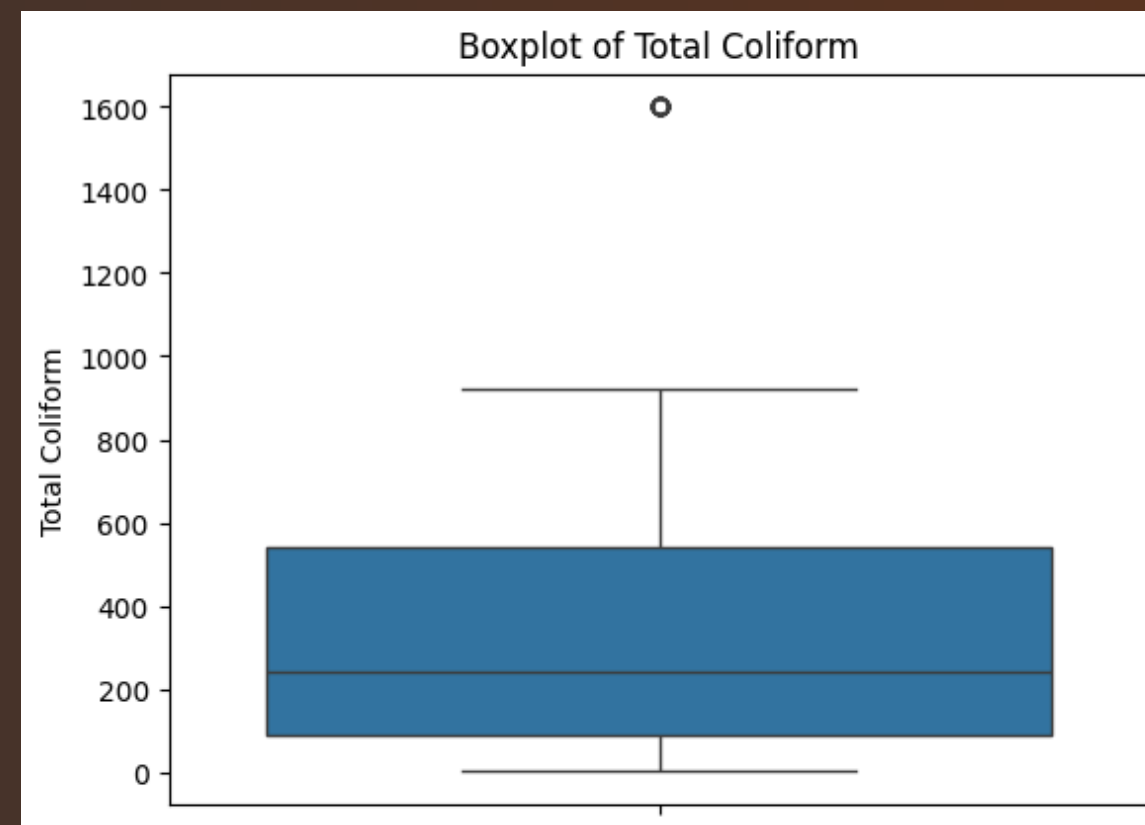


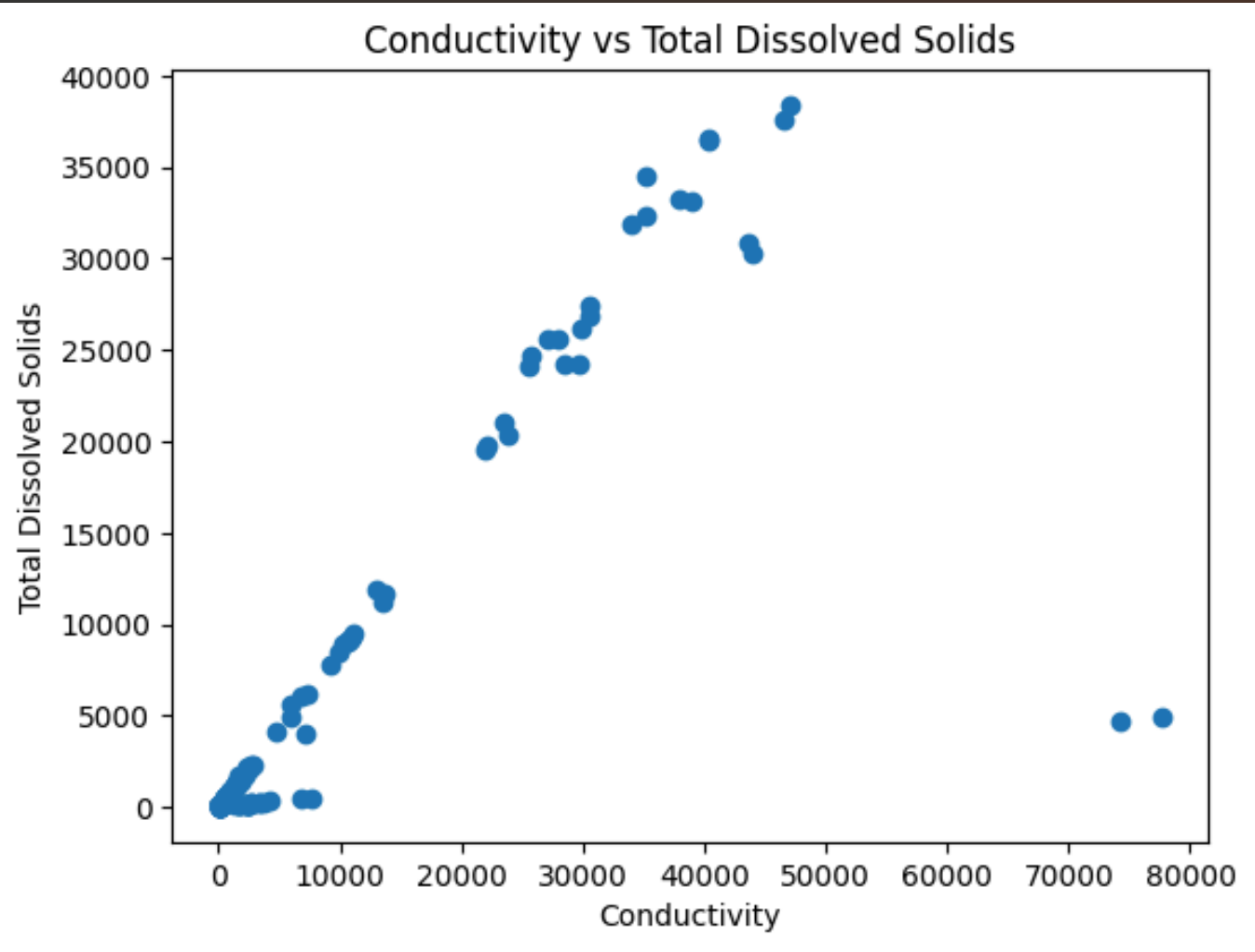
- **COD is highly right-skewed**
- **To examine whether dissolved oxygen follows a normal pattern and detect unusual values.(4 and 7 mg/L)**
- **central tendency and normality of pH values.**



Biochemical Oxygen Demand (BOD) varies across different water use classes and evaluate its ability to differentiate between classes

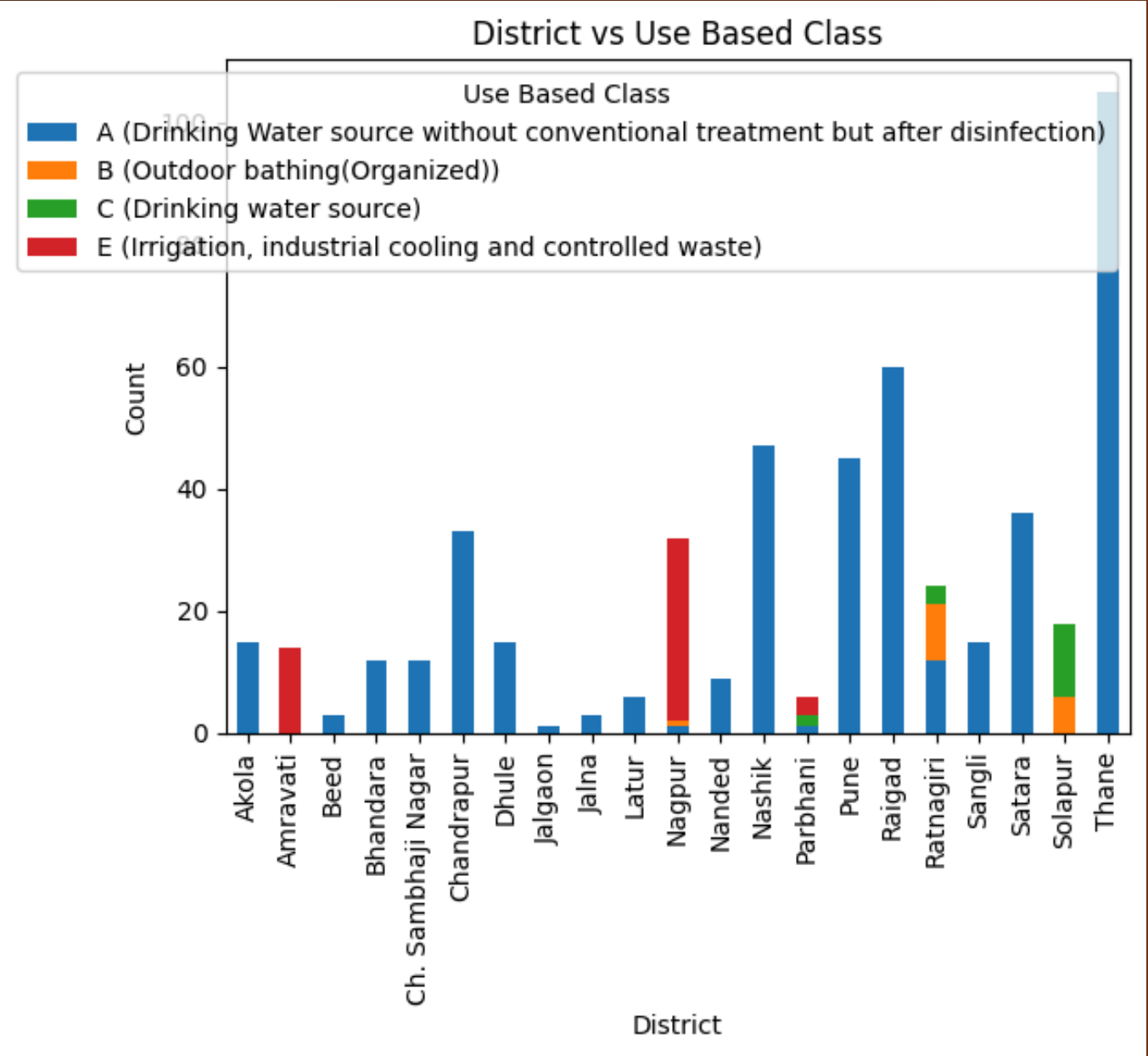
Fecal coliform values are highly right-skewed with several large outliers, suggesting strong variability and need for transformation or capping.





strong positive linear relationship between conductivity and total dissolved solids, suggesting that one of these features could be dropped to reduce multicollinearity

Some districts have a higher concentration of lower-quality classes compared to others, indicating location plays an important role in determining water quality and can be a useful predictive feature.



- Domain Expertise: Prioritized parameters reflecting organic pollution (BOD/COD), oxygen levels, and microbial contamination.
- Correlation Filtering: Dropped redundant variables to mitigate multicollinearity (e.g., retained Conductivity and TDS; dropped Sodium and TSS).
- EDA Insights: Retained BOD, COD, and pH due to high variance across water classes; included Type Water Body as a critical environmental context.

Core Parameters:

- Chemical/Physical: pH, Dissolved O₂, BOD, COD, Turbidity, Temp, Hardness.
- Nutrients/Salinity: Nitrate N, Phosphate, Chlorides, Conductivity, TDS.
- Microbiological: Total & Fecal Coliform.

Model	Train F1 Score	Test F1 Score
SVM (Scaled)	0.6809	0.6799
Logistic Regression (Scaled)	0.5640	0.5690
Random Forest (Tuned)	0.9918	0.8425
LightGBM (Tuned)	1.0000	0.8750

